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GAME THEORETIC BEST-RESPONSE DYNAMICS FOR EVACUEES' EXIT SELECTION

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We present a model for evacuees' exit selection in emergency evacuations. The model is based on the game theoretic concept of best-response dynamics, where each player updates his strategy periodically by reacting optimally to other players' strategies. A fixed point of the system of all players' best-response functions defines a Nash equilibrium (NE) of the game. In the model, the players are the evacuees and the strategies are the possible target exits. We present a mathematical formulation for the model and show that the game has a NE with pure strategies. We also analyze different iterative methods for finding the NE and derive an upper bound for the number of iterations needed to find the equilibrium. Numerical simulations are used to analyze the properties of the model.

Keywords: Evacuation simulation; best-response dynamics; exit selection; agent-based modeling; Nash equilibria.

1. Introduction

The evacuation of buildings containing large crowds is a complex process where many factors may affect the final outcome. Today, the computational evacuation models are commonly used as tools of safety engineering, and fire safety engineering in particular, to predict the evacuation patterns in buildings and transport vehicles, such as passenger ships. Typically, the evacuation process consists of detection and alarm phase, reaction phase and finally the actual movement towards the exits. When modeling evacuees' behavior on individual level, it is essential to consider the behavioral aspects that may change the course of the evacuation. One of the most vital parts of evacuees' decisions is the selection of the exit route. In some of the current evacuation models, the exit selection is modeled simply by allocating

each agent to the nearest exit. Also many prescriptive fire codes implicitly assume that the total exit capacity is used in evacuation. According to experience and studies, these assumptions are unrealistic in many occasions [23, 22]. Factors such as the familiarity of the exit, or congestion in front of the exit may greatly affect the decisions.

During a fire evacuation, an exit that is closer than the others may become unattractive due to heavy congestion. On the other hand, there may be exits that are nearby and unoccupied but remain unused by evacuees. This is because people may be unaware of these exits or avoid them because they have not used them before [25], which is often the case with emergency exits [22]. Another aspect to be taken into account, especially in fire evacuations, are the conditions on different exit routes. Fire and smoke may block some exits and cause completely new situations. To take these behavioral aspects into account in a computational evacuation model, the agents need to have some intelligence. When the intelligence is built inside the agents of an evacuation software, the user is not required to determine the actions of every agent in the input when running simulations.

In this article, we present a game theoretic model for the evacuees' exit selection process. It is based on the concept of *best-response dynamics*. Similar approach has been used, e.g. to study traffic flow in telecommunications networks [1, 15]. In the model, the agents select the target exit, through which they estimate to evacuate the fastest. The actions of other agents may affect the evacuation times, and thus should be taken into account in the decision. In the model, the evacuees select the exit, which is their optimal response to the other agents' actions. The applicability of the game theoretic approach in exit selection depends on the mathematical properties of the system. It is important to know if the system behaves in a chaotic manner, in which case a purely probabilistic treatment may be the most adequate, or if consistent and repeatable solutions can be expected. Mathematically, this question can be answered by studying the existence of the equilibrium and the rate at which the system converges towards this equilibrium. We show that under suitable simplifying assumptions, the game has a Nash equilibrium (NE) with pure strategies. We present different decentralized algorithms that can be used to update the best-response dynamics and show that these algorithms converge to a NE. An upper limit for the number of iterations needed to achieve the equilibrium is also derived. In a preliminary version of this article, a fixed point algorithm without any mathematical analysis was considered [3].

In our analysis, the exit is selected by considering the estimated evacuation times through different exits. This model takes into account the agents' distances to the exits, the amount of crowd between the agent and the exit, and the capacity of the exit. In reality, there are also many other factors affecting the selection. Under some circumstances, the fire-related conditions in the building and the familiarity and visibility of different exits may affect the outcome. We will also present methods for taking these factors into account. The model has been implemented in the FDS + Evac software [13, 12, 14], which combines evacuation simulation with a

state-of-the-art fire-simulation software FDS (Fire Dynamics Simulator) [20, 19]. This program enables the evacuation module to access the data from fire-simulations so that the effect of fire conditions can be explicitly considered in the evacuation models.

Evacuees' exit selection has been previously considered by some authors. The buildingEXODUS software uses an adaptive decision making model [7], where the considered factors include the visibility of the exits and the length of queues at the exits. The effect of fire conditions and exit familiarity on exit selection is also taken into account in the model [8]. The modeling approach of buildingEXODUS is heuristic and the underlying formulas and parameters have not been explicitly presented in the articles. Lo *et al.* [18] presented a game theoretic approach for exit selection. The exit selection is modeled as a two-stage process. First, a two-player zero sum game is formed between the crowd and a "virtual entity," and the mixed strategy NE of this game is interpreted as an optimal distribution of agents over the exits. Second, each evacuee is allocated to an exit based on its location and the mixed strategy equilibrium. This model is not an agent-based model, where decisions are made by autonomous agents. Rather, the exit selection of an individual agent is based on centralized computation of the NE and allocation to the exits is done accordingly. When modeling the actions of individual evacuees, this sort of approach cannot be considered very realistic. NE approaches with decentralized agents have also been applied to evacuation contexts [16, 17]. These papers focus on large-scale evacuations of cities or regions and discuss the usability of the equilibrium solution. However, the model is different from our building-scale approach and the model and its properties are not presented and analyzed in detail.

This paper is organized as follows: in Sec. 2, we define formally the exit selection model as an N-player normal form game. In Sec. 3, we show that the game has an NE with pure strategies and in Sec. 4, we consider different decentralized algorithms for the computation of NE and present some convergence analysis. In Sec. 5, we describe the model's practical implementation and extensions. In Sec. 6, we show numerical results of test simulations. Interpretation of the model and results are further discussed in the final section.

2. Game Theoretic Formulation of the Model

In emergency evacuations, congestion at the bottlenecks of exit routes may have a severe impact on the evacuation times. Hence, the attractiveness of an exit route for an evacuee depends essentially on the decisions of the other evacuees. The mathematical method for modeling such interdependent decisions is game theory.

In our model, the agents are assumed to update their strategies based on their best-response functions in a myopic manner. A fixed point of the system of these functions is a NE of the game. Best-response dynamics have been successfully used in many fields of science, e.g. to stabilize flows in telecommunications networks [1, 15].

To formulate our exit selection model as an *N-player game* in normal form, we begin by defining the concepts of best-response function and NE. For thorough explanation of the concepts, see Ref. 5.

2.1. *An N-player game*

Normal form static game of N players is specified by the players,' or agents,' *strategy spaces* $S_1, \ldots, S_N$, here assumed finite sets, and *payoff functions* $u_1, \ldots, u_N$. This game is denoted by $G = \{S_1, \ldots, S_N; u_1, \ldots, u_N\}$. Function $u_i(s_1, \ldots, s_N)$ defines the payoff to player i if the players choose strategies $(s_1, \ldots, s_N)$, where $s_i \in S_i$, $\forall\, i$. The objective of each player is to select the strategy, which maximizes his own payoff, given that also other players maximize their payoffs. In an implementation of this one-stage game the players act according to their maximizing strategies.

The *best-response function* of player i is defined by

$$s_i := BR_i(s_{-i}) := \arg\max_{s_i' \in S_i} u_i(s_i', s_{-i}), \tag{1}$$

where $s_{-i} := (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_N)$. This function defines the strategy that maximizes the payoff of player i when the other players play s_{-i}.

A NE of the game is a profile of strategies $s^* = (s_1^*, \ldots, s_N^*)$ such that for each player i, strategy s_i^* is the best response to the strategies specified for the other $N-1$ players $(s_1^*, \ldots, s_{i-1}^*, s_{i+1}^*, \ldots, s_N^*)$:

$$s_i^* = \arg\max_{s_i \in S_i} u_i(s_1^*, \ldots, s_{i-1}^*, s_i, s_{i+1}^*, \ldots, s_N^*), \quad \forall\, i. \tag{2}$$

A game may not have a NE in *pure strategies*, but in *mixed strategies*, i.e. when the strategies are distributions over the sets of pure strategies S_i, any game has at least one equilibrium. This was shown by John Nash in 1950 [21].

The best-response function is also called *best-response correspondence*, since in the case $BR(s_{-i})$ is not unique, it is defined to be set valued, and for given s_{-i} we may write $s_i \in BR(s_{-i})$. If a strategy profile $\bar{s} = (\bar{s}_1, \ldots, \bar{s}_n)$ satisfies the equation

$$\bar{s}_i = BR_i(\bar{s}_{-i}), \quad \forall\, i, \tag{3}$$

then, by definition, $\bar{s}$ is a NE of the game. Mathematically, note that $\bar{s}$ is a *fixed point* of the system of all players' best-response correspondences.

An iterative process, where the players update their strategies according to their best responses to the current strategy profile will under certain assumptions converge to a NE [5]. In this paper, we shall formulate the evacuees' exit selection as an N-player game, and interpret the best-response iteration as an adaptive process describing the exit selection dynamics. We shall also prove that the exit selection game indeed has an NE with pure strategies.

2.2. *Exit selection process*

The main goal of an agent is to maximize its individual payoff. In game theory, a player's payoff does not only depend on his strategy, but also on the strategies of the other players. In our model, the main goal of the agents is to escape from the

burning building as soon as possible. Hence, their payoff functions, or from now on called *cost functions*, to be minimized are their evacuation times.

To calculate an estimate for an agent's evacuation time through an exit, one needs to consider two things: the distance to the exit and the possible retarding effect of congestion in front of the exit. Thus, the *estimated evacuation time* of an agent is calculated as the sum of *estimated moving time* and *estimated queuing time*. The moving time is estimated simply by dividing the Euclidean distance to the exit by the unimpeded walking speed of the agent. Similar cost functions have been used in the studies of telecommunication and road traffic networks [2].

The queuing time of an agent at an exit depends on the capacity of the exit and on the number of the other agents that are heading to that exit and are closer to it than the agent himself. Adding the queuing time into the model in a fashion where the queuing time of an agent depends not only on the locations of other agents but also on their target exits, makes the decision of an agent dependent on the decisions of the others. This makes our model a genuine game model.

We analyze the exit selection game assuming that the locations of the agents are fixed. Similarly, the analysis of the best-response algorithms presented in Sec. 4 is performed with the same assumption. Nevertheless, during an evacuation the agents are moving. We will discuss the online-implementation of the best-response algorithms to such situations in Secs. 5 and 7.

This basic model describes the actions of agents in simple exit selection situations. However, an evacuation situation may have many other attributes that affect the decisions. Smoke may block the use of some exits, or agents may be unfamiliar with them and thus prefer other exits. Evacuees may also be keen to stick to the current plan, even if better options become available [22]. We consider methods for modeling these effects in Sec. 5.2. Below we present an exact mathematical formulation for the basic model, where the sum of queuing time and walking time is minimized.

2.3. *Formulation of the exit selection game*

We refer to the agents with indices i and j, where $i, j \in \mathcal{N} = \{1, 2, \ldots, N\}$. The strategies of the agents are the exits e_k, $k \in \mathcal{K} = \{1, 2, \ldots, K\}$. Hence, for strategies s_i and strategy sets S_i we have $s_i \in \{e_1, \ldots, e_K\} := S_i$, $i \in \mathcal{N}$. We denote the profile of all agents' strategies by

$$s := (s_1, \ldots, s_N) \in S_1 \times \cdots \times S_N := S, \tag{4}$$

and will also use notation $s_{-i} := (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_N) \in S_{-i}$ for the strategies of all other agents but agent i. The notation (s_i, s_{-i}) means the strategy sequence $s = (s_1, \ldots, s_N)$.

Let us denote the locations of agent i and exit e_k by $\mathbf{r}_i$ and $\mathbf{b}_k$, respectively, $i \in \mathcal{N}$, $k \in \mathcal{K}$, and let $\mathbf{r} := (\mathbf{r}_1, \ldots, \mathbf{r}_N)$. Agent i's distance to exit e_k is

$$d(e_k; \mathbf{r}_i) = ||\mathbf{r}_i - \mathbf{b}_k||. \tag{5}$$

Now, the payoff function of agent i is the estimated time of evacuation, $T_i(s_i, s_{-i}; \mathbf{r})$, which he minimizes. It is the sum of estimated queuing time and estimated moving time. When agent i chooses strategy $s_i = e_k$, T_i is evaluated as

$$T_i(e_k, s_{-i}; \mathbf{r}) = \beta_k \lambda_i(e_k, s_{-i}; \mathbf{r}) + \tau_i(e_k; \mathbf{r}_i), \quad s_{-i} \in S_{-i}, \tag{6}$$

where β_k is a scalar describing the capacity of exit e_k, $\lambda_i(e_k, s_{-i}; \mathbf{r})$ is the number of other agents that are heading to the same exit e_k as agent i and are closer to it, and $\tau_i(e_k; \mathbf{r}_i)$ is the estimated moving time of agent i to exit e_k. The function λ_i is defined by $\lambda_i(e_k, s_{-i}; \mathbf{r}) := |\Lambda_i(e_k, s_{-i}; \mathbf{r})|$, where

$$\Lambda_i(e_k, s_{-i}; \mathbf{r}) := \{j \in \mathcal{N} | s_j = e_k \,,\, d(e_k; \mathbf{r}_j) < d(e_k; \mathbf{r}_i)\}, \tag{7}$$

and $|\cdot|$ denotes the number of elements in a subset of N. The estimated moving time to an exit is calculated by

$$\tau_i(e_k; \mathbf{r}_i) := \frac{1}{v_i^0} d(e_k; \mathbf{r}_i), \tag{8}$$

where v_i^0 is the moving speed of agent i.

The estimated evacuation times calculated by the agents are not exact predictions of the future but rather estimates by agents with limited intelligence. The definition of Λ_i does not consider the speeds of the other agents, and thus, agents are not able to predict possible overtaking. Also, to precisely calculate the estimated evacuation time, the agents would need to exactly predict the dynamics of the upcoming evacuation. However, when modeling the behavior of evacuees, it is not reasonable to assume such highly intelligent agents. In this model, the abilities of the agents are on a level that is realistic for evacuating people.

3. Existence of a Nash Equilibrium

In this section we show that the exit selection game, described in the previous section, has a NE in pure strategies provided that $\mathbf{r}$ is fixed and the walking speeds are equal for all agents. The existence of a pure strategy equilibrium is a convenient affair, since general existence theorems only imply that an N-player game has an NE with mixed strategies [5].

Theorem 3.1. *Suppose $v_i^0 = v^0$ for all i. Then the exit selection game has a Nash equilibrium in pure strategies.*

Proof. We prove the result by induction with respect to $i \in \mathcal{N}$, by fixing one agent's equilibrium strategy at a time. Suppose, the locations of all agents are given in vector $\mathbf{r}$. Pick up $i_1 \in \mathcal{N}$, $k_1 \in \mathcal{K}$ such that

$$\tau_{i_1}(e_{k_1}; \mathbf{r}_{i_1}) \leq \tau_i(e_k; \mathbf{r}_i), \quad \forall\, i \in \mathcal{N}, \quad \forall\, k \in \mathcal{K}. \tag{9}$$

Since $v_i^0 = v^0$, $\forall\, i \in \mathcal{N}$, (8) and (9) imply that $d(e_{k_1}; \mathbf{r}_{i_1}) \leq d(e_k; \mathbf{r}_i)$, $\forall\, i \in \mathcal{N}$, $\forall\, k \in \mathcal{K}$. In particular this holds for $k = k_1$, so that using (7) we get

$$\lambda_{i_1}(e_{k_1}, s_{-i_1}; \mathbf{r}) = 0 \leq \lambda_{i_1}(e_k, s_{-i_1}; \mathbf{r}), \quad \forall\, s_{-i_1} \in S_{-i_1}, \quad \forall\, k \in \mathcal{K}. \tag{10}$$

Hence, (9) and (10) imply

$$T_{i_1}(e_{k_1}, s_{-i_1}; \mathbf{r}) \leq T_{i_1}(e_k, s_{-i_1}; \mathbf{r}), \quad \forall\, s_{-i_1} \in S_{-i_1}, \quad \forall\, k \in \mathcal{K}, \tag{11}$$

and e_{k_1} is the best response of agent i_1 to every strategy combination of the opponents. Denote $\bar{s}_{i_1} := e_{k_1}$, and fix the strategy of i_1 to be $\bar{s}_{i_1}$ to the end of the process. Note that, from now on $S_{i_1} = \{\bar{s}_{i_1}\}$, and this also holds when we consider product strategy spaces S, S_{-i}, etc.

Let us divide the set of all agents $\mathcal{N}$ into two sets: $\mathcal{N}^F := \{i_1\}$ and $\mathcal{N}^U := \mathcal{N} \setminus \mathcal{N}^F$. In what follows, set $\mathcal{N}^F$ will contain the agents j whose strategies s_j have been fixed by the process so far, say s_j is fixed to $\bar{s}_j$; and set $\mathcal{N}^U$ will contain the agents the strategies of which have not been fixed. We also define two queues for each agent i at each exit e_k, one created by the agents in $\mathcal{N}^F$ and the other by the agents in $\mathcal{N}^U$:

$$\lambda_i^F(e_k; \mathbf{r}_i) := |\{j \in \mathcal{N}^F | \bar{s}_j = e_k,\, d(e_k; \mathbf{r}_j) < d(e_k; \mathbf{r}_i)\}|, \tag{12}$$

$$\lambda_i^U(e_k, s_{-i}; \mathbf{r}) := |\{j \in \mathcal{N}^U | s_j = e_k,\, d(e_k; \mathbf{r}_j) < d(e_k; \mathbf{r}_i)\}|. \tag{13}$$

That λ_i^F does not explicitly depend on s_{-i}, whereas λ_i^F does so, reflects the fact that only the strategies of the agents in $\mathcal{N}^U$ can still be varied. Then the evacuation time of agent i through exit e_k can be written as

$$T_i(e_k, s_{-i}; \mathbf{r}) = \beta_k \lambda_i^U(e_k, s_{-i}; \mathbf{r}) + \beta_k \lambda_i^F(e_k; \mathbf{r}_i) + \tau_i(e_k; \mathbf{r}_i). \tag{14}$$

Now suppose $n \geq 2$ and the process has been repeated $n-1$ times. The strategies for agents $i_1, \ldots, i_{n-1}$ have been fixed to $\bar{s}_{i_1}, \ldots, \bar{s}_{i_{n-1}}$. Also, $\mathcal{N}^F = \{i_1, \ldots, i_{n-1}\}$, $\mathcal{N}^U$, S and the functions λ_i^F and λ_i^U have been updated accordingly. Then at step n of the process, pick up $i_n \in \mathcal{N}^U$, $k_n \in \mathcal{K}$ such that

$$\beta_{k_n} \lambda_{i_n}^F(e_{k_n}; \mathbf{r}_{i_n}) + \tau_{i_n}(e_{k_n}; \mathbf{r}_{i_n}) \leq \beta_k \lambda_i^F(e_k; \mathbf{r}_i) + \tau_i(e_k; \mathbf{r}_i), \tag{15}$$

$\forall\, i \in \mathcal{N}^U$, $\forall\, k \in \mathcal{K}$. Then, (15) implies that

$$d(e_{k_n}; \mathbf{r}_{i_n}) \leq d(e_{k_n}; \mathbf{r}_i), \quad \forall\, i \in \mathcal{N}^U. \tag{16}$$

To show this, suppose (16) does not hold. Then, $\exists i \in \mathcal{N}^U$ such that $d(e_{k_n}; \mathbf{r}_i) < d(e_{k_n}; \mathbf{r}_{i_n})$, i.e. $\tau_i(e_{k_n}; \mathbf{r}_i) < \tau_{i_n}(e_{k_n}; \mathbf{r}_{i_n})$. Further, by the definition of λ_i^F, we get $\lambda_i^F(e_{k_n}; \mathbf{r}_i) \leq \lambda_{i_n}^F(e_{k_n}; \mathbf{r}_{i_n})$; i.e. since agent i is closer to exit e_{k_n} than agent i_n, the number of fixed agents in front of him and heading to e_{k_n} cannot be larger than that for i_n. This contradicts (15).

Now, (13) and (16) imply that

$$\lambda_{i_n}^U(e_{k_n}, s_{-i_n}; \mathbf{r}) = 0 \leq \lambda_{i_n}^U(e_k, s_{-i_n}; \mathbf{r}), \quad \forall\, s_{-i_n} \in S_{-i_n}, \quad \forall\, k \in \mathcal{K}, \tag{17}$$

and, using (14), (15), and (17) we get

$$T_{i_n}(e_{k_n}, s_{-i_n}; \mathbf{r}) \leq T_{i_n}(e_k, s_{-i_n}; \mathbf{r}), \quad \forall\, s_{-i_n} \in S_{-i_n}, \quad \forall\, k \in \mathcal{K}. \tag{18}$$

Thus, e_{k_n} is the best response of agent i_n to every strategy combination of the opponents. We fix $\bar{s}_{i_n} = e_{k_n}$ till the end of the process, define $\mathcal{N}^F = \{i_1, \ldots, i_n\}$, update $\mathcal{N}^U$, S, λ_i^F, and λ_i^U, and repeat the process.

After N steps, and after reindexing, we should have a strategy profile $\bar{s} = (\bar{s}_1, \ldots, \bar{s}_N)$ with the property

$$T_i(\bar{s}_i, \bar{s}_{-i}; \mathbf{r}) \leq T_i(s_i, \bar{s}_{-i}; \mathbf{r}), \quad \forall\, s_i \in S_i, \quad \forall\, i \in \mathcal{N}. \tag{19}$$

This is equivalent to

$$\bar{s}_i = BR_i(\bar{s}_{-i}; \mathbf{r}), \quad \forall\, i \in \mathcal{N}. \tag{20}$$

By definition, $\bar{s}$ is a NE of the game. □

In Theorem 3.1, we show that the exit selection game has a NE with pure strategies. However, the NE may not be unique. Below, a necessary and sufficient condition for the uniqueness of a given NE is derived.

Corollary 3.1. *Let $\bar{s}$ be a Nash equilibrium of the exit selection game. Then, inequality*

$$T_i(\bar{s}_i, \bar{s}_{-i}; \mathbf{r}) < T_i(s_i, \bar{s}_{-i}; \mathbf{r}), \quad \forall\, s_i \neq \bar{s}_i, \quad s_i \in S_i, \quad \forall\, i \in \mathcal{N}, \tag{21}$$

holds if and only if inequality (18) *holds strictly for $k \neq k_n$ on each step of the process described in the proof of Theorem* 3.1.

Proof. Suppose inequality (18) holds strictly on each step of the process. Then, on step n of the process, e_{k_n} is the strict best response of agent i_n, $\forall\, s_{-i_n} \in S_{-i_n}$. Recall that in S_{-i_n} the strategies of agents $i_1, \ldots, i_{n-1}$ have been fixed to $\bar{s}_{i_1}, \ldots, \bar{s}_{i_{n-1}}$, respectively, at the earlier steps of the process, but the strategies of the other agents can be arbitrary. Especially, these strategies can be chosen to be the ones selected by the remaining steps of the process that produces the equilibrium $\bar{s}$. Hence, inequality (21) holds.

Now suppose inequality (21) holds, but suppose (18) does not hold strictly at some step n of the process. Then, at some step n it holds:

$$T_{i_n}(e_{k_n}, s_{-i_n}; \mathbf{r}) = T_{i_n}(e_{k'_n}, s_{-i_n}; \mathbf{r}), \tag{22}$$

for some $s_{-i_n} \in S_{-i_n}$, and for some $k'_n \neq k_n$, $k'_n \in \mathcal{K}$. Now, because agent i_n and exit e_{k_n} satisfy inequalities (17) and (15) at step n, also agent i_n and exit e_{k_n} replaced by $e_{k'_n}$ do so. Hence, as shown in the proof of Theorem 3.1,

$$T_{i_n}(e_{k'_n}, s_{-i_n}; \mathbf{r}) \leq T_{i_n}(e_k, s_{-i_n}; \mathbf{r}), \quad \forall\, s_{-i_n} \in S_{-i_n}, \quad \forall\, k \in \mathcal{K}, \tag{23}$$

and thus, Eq. (22) does not only hold for some particular strategies of the other agents, but also $\forall\, s_{-i_n} \in S_{-i_n}$. Especially it holds with the strategies chosen in the remaining steps of the process. This contradicts with (21), and thus, if (21) holds, inequality (18) holds strictly on each step of the process. □

Corollary 3.2. *Inequality* (21) *is a necessary and sufficient condition for the uniqueness of a NE $\bar{s}$.*

Proof. From literature we know that if the process of *iterated elimination of strictly dominated strategies* eliminates all but the strategies $(\bar{s}_1, \ldots, \bar{s}_N)$, then these strategies are the unique NE of the game [6]. The process described in the proof of Theorem 3.1 defines exactly such a process if and only if inequality (18) holds strictly on each round. This observation together with Corollary 3.1 complete the proof. □

Inequality (21) gives a convenient way to check whether a computed equilibrium is unique or not.

Here we have considered the case where the cost function of an agent is the sum of estimated queuing time and estimated moving time. Nevertheless, the presented analysis holds even if we consider more general functions, e.g. arbitrary weighted sum of these two quantities, the maximum of them, or an arbitrary increasing function of them.

In Theorem 3.1, we make the assumption that the walking speeds of all agents are equal. When modeling the dynamics of large crowds, this assumption is not very limiting because overtaking can be difficult and many people adjust their speed to the flow. However, it would be possible to alter the model so that all agents would know each others' walking speeds and be able to predict overtaking. Actually, a pure strategy NE exists also for this model. The proof for this is very similar to the proof of Theorem 3.1 and we will not present it in this paper. Also in this case, the strategies of the agents can be fixed to their equilibrium strategies one by one, in an ascending order of their evacuation times.

It is also interesting to study whether the NE is also an optimal solution of the exit selection problem, i.e. if it minimizes the sum of all agents' estimated evacuation times. Another condition for a solutions' optimality in an evacuation context would be that it minimizes the maximum of all agents' individual evacuation times, i.e. the time when all agents have evacuated. A simple counter example with two agents shows that the NE is not the solution for these problems: Assume two agents i and j, and two exits e_1 and e_2. Suppose the walking times to the exits are 1 and 2 s respectively, for agent i and 2 and 5 s for agent j and suppose the capacity of both exits is one person in 2 s. Now the NE of the game is (e_1, e_1) since the best response of both players is e_1 regardless of the other agent's strategy. However, the sum and maximum of the estimated evacuation times with the NE are 5 and 4 s, respectively, while strategy profile (e_2, e_1) results in sum of 4 s and maximum of 2 s. Hence, each agent minimizing its individual evacuation time does not produce the optimal outcome for the whole crowd. The agents only minimize their individual evacuation times, and thus, are unable to give way to other agents by selecting a slightly slower exit.

4. Decentralized Algorithms

From the above theorem, we know that the presented exit selection game does have a NE. Nevertheless, as we are modeling populations of independent agents, we wish

to know if this equilibrium can be achieved without communication or coordination, i.e. by *decentralized algorithms*. In decentralized algorithms, the agents make their decisions by only observing the actions of the other agents and update these decisions in an *online* fashion. *Common knowledge* of the payoff functions of the agents is not required.

In this section, we present some decentralized best-response algorithms and study their convergence. We show that, irrespective of the large search space of N^K strategy combinations, these algorithms converge to a NE very fast, the upper bound of the number of iterations being N. The agents update strategies which take values on discrete sets. In such setting, the best-response updating may lead to cycles of different forms, see Sec. 1.2.5 in Ref. 5, which makes the convergence of these algorithms an interesting result. It should also be noted that the best-response algorithms considered here do not take into account the past or possible forthcoming updates but are myopic in their nature.

A typical best-response algorithm will have the form

$$s_i^{(t)} = \begin{cases} BR_i(s_{-i}^{(t-1)}; \mathbf{r}), & i \in \mathcal{N}_t \\ s_i^{(t-1)}, & \text{otherwise}, \end{cases} \tag{24}$$

where $s_i^{(t)}$ denotes the strategy of agent i on iteration period t. The difference between the algorithms is in the choice of the set $\mathcal{N}_t \subset \mathcal{N}$. Here we identify three possible choices following [1]:

(1) *Parallel Update Algorithm* (PUA): $\mathcal{N}_t = \{1, \ldots, N\} = \mathcal{N}$ for all t.
(2) *Round Robin Algorithm* (RRA): $\mathcal{N}_t$ contains one agent $i \in \mathcal{N}$ in each period. The updating order is an arbitrary permutation of the index set that is repeated. Hence, in every subsequent N periods each agent updates its strategy exactly once.
(3) *Random Polling Algorithm* (RPA): $\mathcal{N}_t = \{\xi_t\}$, where ξ_t follows a discrete uniform distribution over the set of agents $\{1, \ldots, N\}$ for all t.

We present numerical experiments using these algorithms in Sec. 6. Below we study their convergence.

Theorem 4.1. *The Parallel Update Algorithm* (*PUA*) *converges to a Nash equilibrium in at most N iterations, where N is the number of agents.*

Proof. Assume $\mathbf{r}$ defines the locations of the agents and their initial target exits are defined by $s^{(0)} \in S$. From the proof of Theorem 4.1, we know that in period 1 there is an agent i_1 and exit e_{k_1} satisfying inequality (11). Thus, exit e_{k_1} is the best response of agent i_1 regardless of the actions of the other agents. In PUA, all agents update their strategies to their best responses in each period. Hence, the

strategy of agent i_1 will be updated in the first period to his equilibrium strategy $\bar{s}_{i_1}$ and remain fixed throughout the process:

$$s_{i_1}^{(t)} = \bar{s}_{i_1} = e_{k_1}, \quad t = 1, 2, \ldots. \tag{25}$$

Now suppose $n \geq 2$ and the PUA has been iterated $n-1$ times, and $n-1$ agents have permanently set their strategies to their equilibrium strategies:

$$(s_{i_1}^{(n-1)}, \ldots, s_{i_{n-1}}^{(n-1)}) = (\bar{s}_{i_1}, \ldots, \bar{s}_{i_{n-1}}). \tag{26}$$

We set $\mathcal{N}^F = \{i_1, \ldots, i_{n-1}\}$, $\mathcal{N}^U = \mathcal{N} \setminus \mathcal{N}^F$, and pick up agent i_n and exit e_{k_n} satisfying (18). Now, on period $t = n$, $\bar{s}_{i_n} = e_{k_n}$ is the best response of agent i_n regardless of the actions of the agents in $\mathcal{N}^U$, and

$$s_{i_n}^{(t)} = \bar{s}_{i_n} = e_{k_n}, \quad t = n, \;\; n+1, \ldots. \tag{27}$$

Using induction, we have shown that in each period at least one agent will permanently update his strategy to an equilibrium strategy. Hence, after N periods, the algorithm has converged to a NE. □

Theorem 4.2. *The Round Robin Algorithm converges to the Nash equilibrium in at most N^2 iterations, where N is the number of agents.*

Proof. Again, assume N agents have locations $\mathbf{r}$ and random initial target exits $s^{(0)}$. As stated in the proof of Theorem 4.2, if agent i_1 and exit e_{k_1} satisfy inequality (11), e_{k_1} is the best response of agent i_1 regardless of the other agents' actions.

In RRA, in every subsequent N periods, each agent updates his strategy exactly once. Hence, an upper bound for agent i_1's first update is N, and we can write:

$$s_{i_1}^{(t)} = \bar{s}_{i_1} = e_{k_1}, \quad t = N, \;\; N+1, \ldots. \tag{28}$$

Now suppose M periods of the RRA have been iterated and $n-1$ agents have permanently set their strategies to an equilibrium strategy:

$$(s_{i_1}^{(M)}, \ldots, s_{i_{n-1}}^{(M)}) = (\bar{s}_{i_1}, \ldots, \bar{s}_{i_{n-1}}). \tag{29}$$

We set $\mathcal{N}^F = \{i_1, \ldots, i_{n-1}\}$, $\mathcal{N}^U = \mathcal{N} \setminus \mathcal{N}^F$, and pick up agent i_n and exit e_{k_n} satisfying (18). In this situation, $\bar{s}_{i_n} = e_{k_n}$ is the best response of agent i_n regardless of the actions of the agents in $\mathcal{N}^U$. The strategy of agent i_n will certainty be updated to e_{k_n} in the upcoming N periods, and thus, we can write:

$$s_{i_n}^{(t)} = \bar{s}_{i_n} = e_{k_n}, \quad t = M+N, \;\; M+N+1, \ldots. \tag{30}$$

Using induction we have shown that in every subsequent N periods at least one new agent will permanently set his strategy to his equilibrium strategy. Hence, with N agents an upper bound for achieving a NE is N^2 periods. □

Remark 4.1. Above we derived upper bounds for the number of iterations with the PUA and RRA. In Sec. 6, we present results of test simulations with these

algorithms and use the term *iteration round* for the round during which all the agents have been updated once. Hence, with the RRA an iteration round means N iterations of the best-response algorithm, while with the PUA it is one iteration. In terms of iteration rounds, the upper bounds of PUA and RRA are equal, N iteration rounds with both algorithms. Also, the computational costs of the two algorithms over an iteration round do not differ significantly.

In addition to the RPA, there are other stochastic algorithms, e.g. *Stochastic Asynchronous Algorithm* (SAA) [1]. In SAA, in each period a randomly picked set of agents update their strategies, where each agent is picked on each round with probability p. No upper bounds can be derived for the convergence of these stochastic algorithms, but computational studies of their properties would be interesting. For instance, one could study how the convergence of SAA depends on the value of parameter p and search for an optimal value in the spirit of Ref. 24. Nevertheless, in this article we focus on the properties of PUA and RRA.

Above we deduced theoretic upper bounds for the number of iterations with different algorithms. In practice, it turns out that in many situations the algorithms converge much faster. In Sec. 6, we study and compare the computational properties of these algorithms.

5. Practical Implementation

In the earlier sections, we have shown that the exit selection game has a NE which can be found with decentralized best-response algorithms. These proofs were based on the assumption that the locations of the agents are fixed. The question now arises if the algorithms can be used to describe the actions of moving agents in a simulation environment in an online fashion. It should be noted that our formulation is not genuinely dynamic, since the model does not take into account the agents' changing locations (this matter is further discussed in Sec. 7). Hence, the NE of the game may change during an evacuation as the agents move or the environment changes.

When applying the model to an evacuation simulation software with moving agents, the possibility of a change in the equilibrium should be considered. Hence, it is not reasonable to assume that agents only calculate the equilibrium at a particular instant, i.e. the initial situation, but they should also be able to update their strategies throughout the evacuation. The analysis of Sec. 4 and the numerical results of Sec. 6 show that the best-response algorithms converge to the NE very fast, and thus it is possible to recalculate the NE on the fly, as the situation changes. For example, each agent could be set to update its strategy frequently throughout the evacuation to make the crowd rapidly react to a changing equilibrium.

5.1. *FDS + Evac*

The exit selection model is implemented in the FDS + Evac evacuation simulation software [12]. The evacuation software is a module on a popular fire-simulation

software FDS [20, 19] and this setting enables the integration of evacuation simulation and state-of-the-art fire-simulation. Hence, the agents reactions to the progress of fire can be modeled dynamically.

In FDS + Evac, the agents move in a continuous space and the movement and physical interactions are modeled with the *social force model* of Helbing *et al.* [11, 9, 10]. The main advantage of the social force model is that its equations are based on the actual physical forces arising in crowds. It also enables the modeling of all kinds of behavior. User can give the agents a desired moving direction and speed, and the agent will try to achieve it, but the motion is restricted by physical restrictions, e.g. inertia, physical contact forces between agents, and a "social force" that describes agents' tendency to keep a little distance to other agents.

5.2. *Additional features*

The exit selection of evacuees may be affected by many other factors than the estimated evacuation time. People tend to use familiar exit routes even if there are faster routes available. According to Proulx [23], evacuees prefer familiar alternatives because they feel that unknown alternatives increase the threat. For the same reason, evacuees can be considered to select visible exits over the invisible ones. The FDS + Evac software enables the evacuation simulator to use the fire-related data of FDS. This makes it possible also to consider the physical conditions, like temperature and smokiness, in the exit selection model. In this section, we present a method for taking into account the visibility, familiarity, and fire conditions in our exit selection model. For a more comprehensive research on evacuees' interaction with fire conditions, see Ref. 8.

In our exit selection model, the familiarity and visibility of the exits and the conditions at the exit are taken into account by constraining the set of feasible exits. These three factors divide the exits into six groups that have a preference order. Each agent will select an exit from the nonempty group that has the highest preference. If there are several exits in this group, the selection between them is made by minimizing the estimated evacuation time as presented in Sec. 2.3.

The effects of *familiarity*, *visibility*, and *fire-related conditions* are taken into account by defining three binary variables

$$\mathrm{fam}_i(e_k), \quad \mathrm{vis}(e_k; \mathbf{r}_i), \quad \mathrm{con}(e_k; \mathbf{r}_i), \quad \forall\, i \in \mathcal{N}, \quad k \in K,$$

where

$$\mathrm{fam}_i(e_k) = \begin{cases} 1, & \text{if exit } e_k \text{ is familiar to agent } i \\ 0, & \text{if exit } e_k \text{ is not familiar to agent } i \end{cases}$$

$$\mathrm{vis}(e_k; \mathbf{r}_i) = \begin{cases} 1, & \text{if exit } e_k \text{ is visible to agent } i \\ 0, & \text{if exit } e_k \text{ is not visible to agent } i \end{cases}$$

$$\mathrm{con}(e_k) = \begin{cases} 1, & \text{if conditions are tolerable at exit } e_k \\ 0, & \text{if conditions are intolerable at exit } e_k. \end{cases}$$

Now the exits can be divided into groups that have preference numbers from one to six. The smaller the preference number is, the more preferable the exit. Definitions for these numbers are presented in Table 1. The order is based on common sense and some social psychological findings. For instance, evacuees prefer familiar routes even if there were faster unfamiliar routes available [23, 22].

Hence, the complete exit selection model can be presented for each agent $i \in \mathcal{N}$ as follows:

$$s_i = BR_i(s_{-i}; \mathbf{r}) = \arg\min_{s_i' \in S_i} T_i(s_i', s_{-i}; \mathbf{r}), \quad \text{st. } s_i' \in E_i(\bar{z}), \tag{31}$$

where $E_i(\bar{z})$ is the nonempty exit group with the best preference number $\bar{z}$ for agent i.

In some situations, an agent may not be able to estimate the queue length in front of an exit. This is especially the case in situations where the agent cannot see the exit. In these cases the estimated evacuation time should not depend on the queuing time, and thus, Eq. (6) can be replaced by

$$T_i(e_k, s_{-i}; \mathbf{r}) = \mathrm{vis}(e_k, \mathbf{r}_i)\beta_k \lambda_i(e_k, s_{-i}; \mathbf{r}) + \tau_i(e_k; \mathbf{r}_i). \tag{32}$$

This makes the estimated evacuation times shorter for the invisible exits. However, this does not affect the functioning of the model, because the estimated evacuation times are only compared between exits in the same exit group.

Sometimes an alternative exit is only slightly faster than the current target exit. We assume that an agent may not be willing to react to such small differences and define an *anchoring parameter* to model this tendency. The parameter describes how much faster an alternative exit needs to be in order for an agent to change its

Table 1. The preference numbers of exit groups used in our model.

Preference number	Exit group	$\mathrm{vis}(e_k, \mathbf{r}_i)$	$\mathrm{fam}_i(e_k)$	$\mathrm{con}(e_k)$
1	$E_i(1)$	1	1	1
2	$E_i(2)$	0	1	1
3	$E_i(3)$	1	0	1
4	$E_i(4)$	1	1	0
5	$E_i(5)$	0	1	0
6	$E_i(6)$	1	0	0
	No preference	0	0	1
	No preference	0	0	0

Note: The smaller the preference number is, the more preferable the exit. The combinations of the last two rows have no preference because the evacuees are unaware of the exits that are unfamiliar and invisible, and thus cannot choose these exits.

target exit. This behavior can be taken into account by subtracting the anchoring parameter from the evacuation time through the current target exit. Another possibility would be to define the anchoring parameter as a proportion of the estimated evacuation time, instead of absolute seconds. In this case, the estimated evacuation time of the current exit is multiplied by the parameter, which can have values between zero and one.

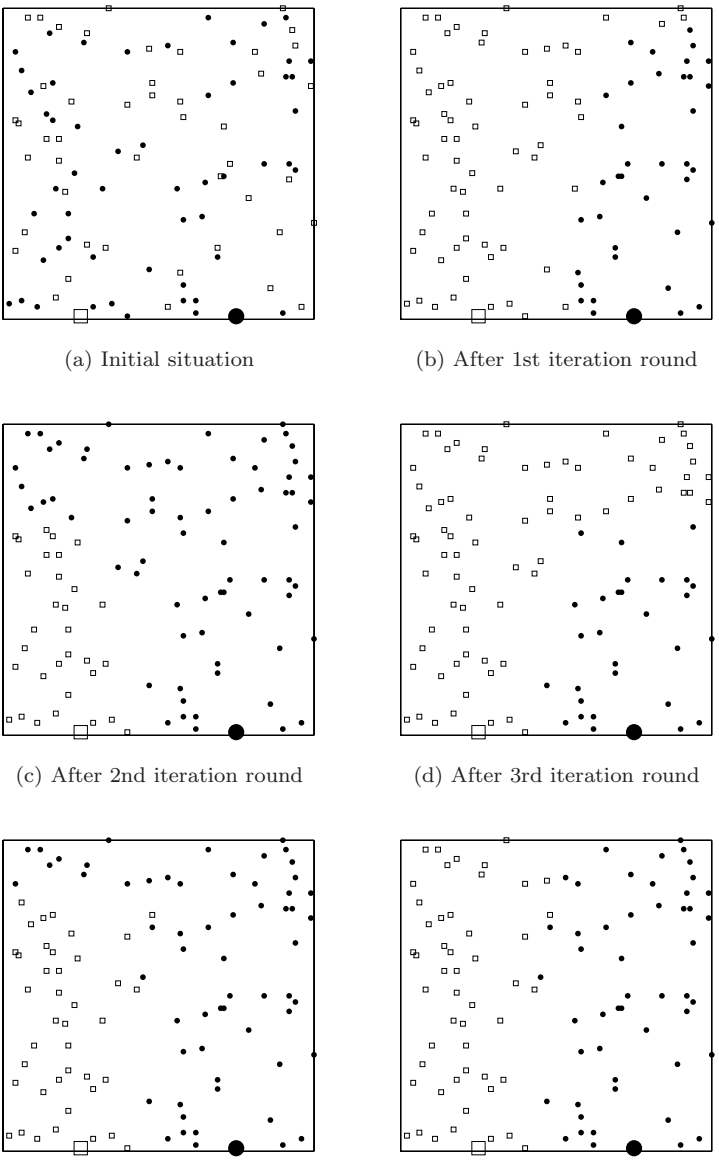
6. Numerical Experiments

The developed exit selection model has been implemented in the FDS + Evac software [12]. FDS + Evac uses the RRA to update the agents' strategies. A simpler simulator was developed to test the convergence of the different update algorithms in simple test geometries without considering the fire-related conditions and the familiarity and visibility of exits. In this section, we compare the convergence properties of PUA and RRA, and determine the effect of some parameters and factors on the convergence.

Test simulations were ran using a simple test geometry and running the two algorithms with the same initial situation. The geometry was a 40 m × 40 m square room with two exits with equal capacity ($\beta_k = 1$ agent per second). Total of 100 agents were randomly located to the room and were given random initial target exits. Figures 1 and 2 illustrate the progress of the two algorithms. It can be seen that both of the algorithms converge to the same equilibrium, but the difference between the needed iteration rounds was very significant. While it took 18 iteration rounds to find the NE with the PUA, the RRA converged after three rounds of updating each agent. The main reason for this seems to be the oscillation that occurs with the PUA; as the agents update their strategies simultaneously, the crowd overreacts to differences in the queuing time. When the queue at one exit is short at some step of the iteration, a big part of the crowd will select that exit on the next step. This increases the queue and usually makes some other exit much faster. This oscillation can be seen in the snapshots of Fig. 1. In the RRA, the agents' strategies are not updated simultaneously and thus, oscillation does not occur.

The algorithm is not the only factor affecting the number of iterations. In general, it can be said that the more sensitive the agents' best responses are to the other agent's actions the more iterations are needed. The factors affecting this sensitivity are agents' walking speed v_0, the capacity of the exits β_k, the number of agents and the building geometry. Also the anchoring parameter has a significant effect.

If the walking speed v_0 of an agent increases, the walking time decreases and the queuing time becomes more important when calculating the best response. This causes the agents' reactions to become more sensitive to the others' actions and increases the number of needed iterations. On the other hand, if the capacities β_k of the exits increase, the queuing time decreases and so does the number of



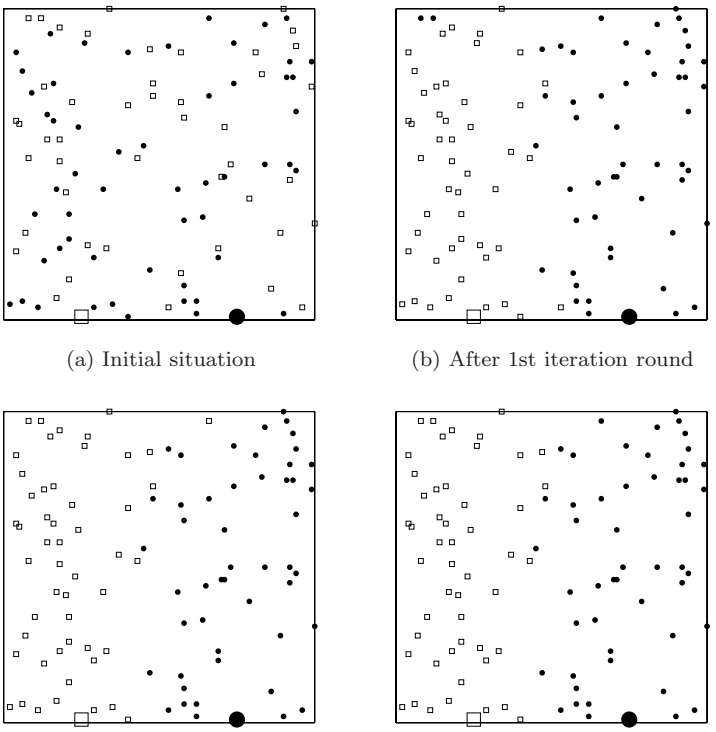
(a) Initial situation (b) After 1st iteration round

(c) After 2nd iteration round (d) After 3rd iteration round

(c) After 10th iteration round (d) After 18th iteration round, equilibrium

Fig. 1. Snapshots of the progress of a PUA-iteration. The exits are marked with the large white square and black circle and the markers of the agents relate to their current target exit.

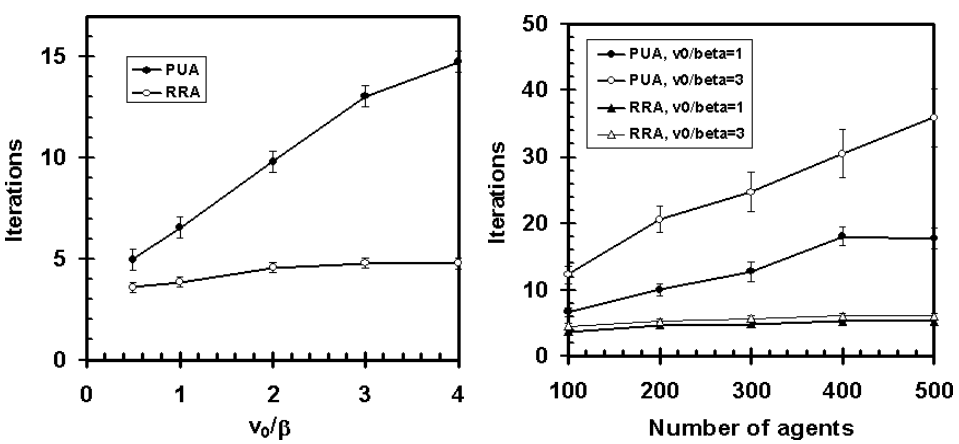
iterations. In test simulations, it turns out that with a certain value of the ratio v_0/β_k the convergence seems to be similar regardless of the individual values of the parameters. Figure 3(a) illustrates the dependence between the value of this ratio and the number of iterations. These simulations were ran with 200 agents in the



(a) Initial situation (b) After 1st iteration round

(c) After 2nd iteration round (d) After 3rd iteration, equilibrium round

Fig. 2. Snapshots of a RRA-iteration.



(a) (b)

Fig. 3. The effects of (a) the ratio v_0/β_k and (b) the number of agents on the number of iteration rounds. The observations of PUA are averages of 40 simulations and the ones with RRA are averages of 20 simulations. The error bars describe a 95% confidence interval for the average number of iterations.

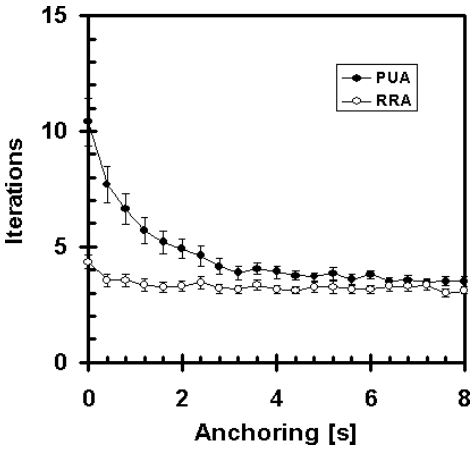


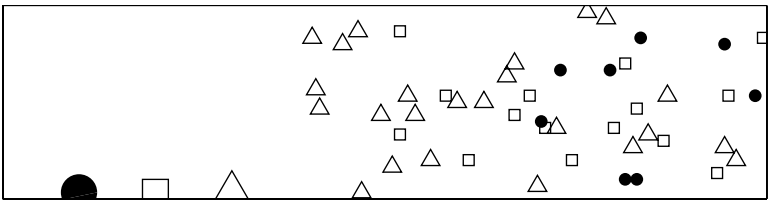
Fig. 4. The average number of iteration rounds versus the value of the anchoring parameter. The observations of PUA are averages of 40 simulations and the ones with RRA are averages of 20 simulations. The error bars describe a 95% confidence interval for the average number of iterations.

test geometry of Figs. 1 and 2. The effect of v_0/β_k is especially significant with the PUA and smaller with the RRA.

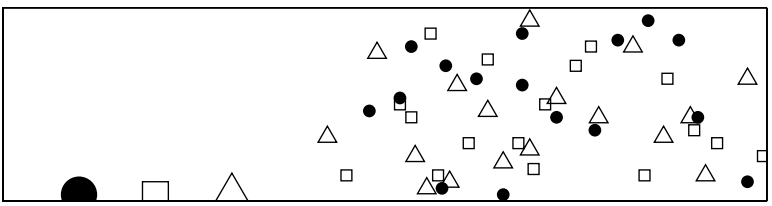
A larger crowd causes a longer queue, and thus increases the queuing time related to the walking time. Figure 3(b) illustrates the dependence between the number of agents and the number of iterations. Also, it is quite natural that increase in the anchoring parameter makes the agents' reactions less sensitive and thus, decreases the number of iterations. According to Fig. 4, this is especially the case with the PUA. Also in this case, these factors mainly affect the number of iterations with the PUA. The RRA finds the equilibrium in around four iteration rounds, regardless of the level of these factors. According to these results, the convergence rate of PUA is very sensitive to the values of v_0/β ratio, crowd size, and anchoring parameter. The RRA in turn finds the equilibrium robustly, regardless of the parameter values.

The building geometry may have a great effect on the speed of convergence. In Sec. 4, we showed that the upper bound for the iterations with PUA is the number of the agents N. In most geometries, the algorithm converges much faster, but there are also situations, where all N iteration rounds are needed. An example of such geometry is given in Fig. 5. The equilibrium showed in Fig. 5 was achieved in 43 iterations, for 50 agents with the value of $v_0/\beta_k = 1$. When the value of v_0/β_k was increased to 20, the maximum number of 50 iterations was needed to achieve the equilibrium. Rather than exit selection, such parameter values could describe the situation of selecting the cashier queue in a supermarket. Also with this geometry, the RRA performs well and finds the equilibrium in just a few iteration rounds.

We also ran numerical experiments using FDS + Evac. The evacuation of 100 agents was simulated in the geometry of Fig. 7. The target exits of the evacuees were



(a) $v_0/\beta_k = 1$



(b) $v_0/\beta_k = 20$

Fig. 5. The equilibria with parameter ratio values (a) $v_0/\beta_k = 1$, (b) $v_0/\beta_k = 20$. The exits are marked with large circle, square, and triangle and the markers of the agents relate to their current target exit.

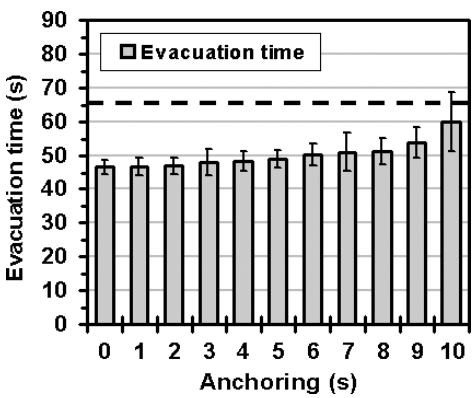


Fig. 6. The total evacuation time of 100 agents with different anchoring parameter values in the geometry of Fig. 7. The dashed line is the evacuation time without the exit selection, i.e. all agents selecting the nearest exit. The observations are averages of 50 simulations and the error bars describe the standard deviations of the evacuation times.

updated so frequently (every 0.01 seconds) that it can be assumed that the system was in equilibrium throughout the simulation. Figure 7 is a snapshot of a simulation with FDS + Evac, and Fig. 6 illustrates the dependence between total evacuation time and anchoring parameter. The dashed line in Fig. 6 is the evacuation time when the agents update their strategies to select the nearest exit, but disregard the queue. This approach is used in many evacuation simulation models. The results indicate that when the exit selection model is used, the 100 agents evacuate the room 29% faster than without the model.

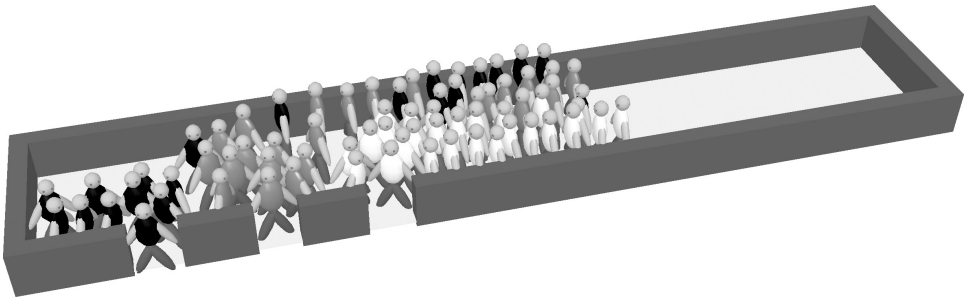


Fig. 7. A snapshot of a test simulation with FDS + Evac. The black agents are heading to the left exit, the gray agents to the middle one, and the white ones to the right exit.

7. Discussion

We present a game theory model for evacuees' exit selection. The goal of each agent is to minimize his estimated evacuation time. We show that the game has a NE in pure strategies and derive a necessary and sufficient condition for its uniqueness. Also, it is shown that the equilibrium can be computed efficiently using decentralized best-response algorithms.

We compute the NE solution of the exit selection game by using following assumptions: The agents' locations $\mathbf{r}$ are constant during the iteration, and every agent updates his current strategy assuming that the other agents will stick to their strategies for future periods. In principle, one could take the changing $\mathbf{r}$ into account by formulating an appropriate state equation for $\mathbf{r}$, then discretizing it with respect to time and space, and using dynamic programming to calculate optimal feedback strategies for the agents. However, employing such strategies requires highly rational agents that take into account the future decisions of all agents until the end of the evacuation. Such abilities are unlikely for the members of a large evacuating crowd. Rather, it is believable that people often behave myopically by reacting to the moves of the crowd. Therefore our model, where agents myopically update their strategies as the environment or the configuration of the crowd changes, could be considered a more realistic approach.

It should be noticed that the existence and convergence results presented in this paper do not only hold for the used cost function, i.e. the sum of the estimated queuing time and the estimated moving time, but for any increasing function of these two quantities, and especially for the function max over the two.

Due to our myopic approach, the best responses of the agents, i.e. the equilibrium, may change when the crowd moves. The agents can react to these changes if they are set to update their strategies frequently during the evacuation. The fast convergence of the best-response algorithms ensure that frequent updating will keep the system close to its current equilibrium throughout the simulation. It should be noted that the behavior obtained by frequently updating the myopic equilibrium is different from the solution of the dynamic model. In the model's implementation to

FDS + Evac, each agent updates his strategy multiple times in a second, and thus, it can be assumed that the system is essentially in equilibrium from start to finish.

The fact that the presented game has a NE in pure strategies is interesting, since general existence theorems only imply that an N-player game has a NE with mixed strategies. The result is important for the justification of our approach, as it would be unrealistic to assume evacuees selecting mixed strategies. Also fast convergence of the best-response algorithms is essential from the applications point of view. If the algorithms would produce cycles or behave in a chaotic manner, the usability of our approach could be questionable. The convergence result is interesting also from a theoretical point of view, since best-response algorithms often tend to create cycles when applied to matrix games with discrete strategy sets [4].

Best-response algorithms are closely related to other learning algorithms in evolutionary game theory, e.g. fictitious play and replicator dynamics [4]. Fictitious play takes into account the frequencies of the past actions and with an appropriate time normalization, discrete fictitious play asymptotically is approximately the same as the continuous time best-response dynamic.

The convergence of the algorithms is tested with numerical simulations. In most cases, the number of iterations needed for convergence is lower than the theoretical upper limit. However, we show that there are situations where the maximum number of iteration rounds is actually needed. In terms of the number of individual updates, the RRA turns out to be much faster than the PUA. The sensitivity studies show that the convergence rate of PUA is very sensitive to the ratio of walking speed and exit flow rate v_0/β, crowd size, and the value of the anchoring parameter. This is mainly because the simultaneous updating of PUA can cause oscillations in the strategies. In all test situations, the RRA converges consistently in around four iteration rounds. For this reason, RRA is used when implementing the model in the FDS + Evac software. The results of FDS + Evac simulations show that in the test geometry the exit selection algorithm presented in this paper significantly reduces the total evacuation time compared to the case where all the agents use the nearest exit.

Acknowledgments

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